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Mid Semester Examination 2025

Name of the Program: B.Tech. Biotechnology

Semester: VI

Name of the Course: Microbial Technology

Course Code: TBT 604c

Time: 90 minutes

Maximum Marks: 50

Note:-

- (i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	What do you understand by the term fermentation? Explain different fermentation process.	
OR		
(b)	Define biopolymers and explain the production of PHA or PHB.	
Q2	(10 marks)	CO3 & CO1
(a)	Implement the knowledge of recombinant DNA technology in production of hGH.	
OR		
(b)	Differentiate between gram positive and gram-negative bacteria with examples.	
Q3	(10 marks)	CO1 & CO2
(a)	Explain Vaccines. Discuss Synthetic and DNA Vaccines.	
OR		
(b)	Discuss the general principle of production of recombinant protein.	
Q4	(10 marks)	CO2
(a)	Draw the well labelled diagram of Hepatitis B virus and give diagrammatic representation of its vaccine production.	
OR		
(b)	Which microorganism is involved in the production of xanthan gum? Explain its production.	
Q5	(10 marks)	CO2
(a)	Explain the production of recombinant insulin.	
OR		
(b)	Discuss the role of microorganisms in human therapeutics, agriculture and environment with examples.	

